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LAB REPORT

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Lab Name : Basic Commands of Linux

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Submitted to
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Title : Basic commands of Linux

Objective:

- To understand the basic linux Command Line Interface (CLI)
- To learn commonly used linux commands
- To perform basic file and directory management.
- To gain familiarity with linux terminal operations.

Introduction:

Linux is a free and open source operating system widely used in software development, networking, cybersecurity, cloude computing and server administration. It provides a powerful 'Command Line Interface (CLI)' that enables user to interact with operating system by executing command.

Linux commands allows users to navigate directories, manage files, monitor system resources, control processes, configure networks and performs administrative tasks efficiently.

Proposed Methodology:

This lab was conducted using Linux Terminal (Git Bash/ Linux Shell). Below various basic linux command discussed accordingly.

1. Command: pwd
   Description: show current directory
   Syntax : pwd

2. Command: ls
   Description: Lists files and directories
   Syntax : ls
3. Command: ls-l
   Description: show files in long format
   Syntax : ls-l

Here is a table of commands and their description:

Commands	Description	Syntax
ls-a	show hidden files	ls -a
ls -lh	Show file size in human readable format	ls -lh
cd	change directory	cd directory_name
mkdir	create directory	mkdir folder_name
rmdir	remove directory	rmdir folder_name
touch	create empty file	touch file.txt
cp	copy files or directory	cp source destination
mv	move or rename	mv old new
rm	delete file	rm file.txt
rm -r	Delete a directory recursivly	rm -r folder-name
cat	display file content	cat file.txt
tac	display file content in revers order	tac file.txt
head	Show first 10 liles of a file	head file.txt
tail	Show last 10 lines of a file	tail file.txt
more	view file page by page	more file.txt
less	view larger file interectively	less file.txt
nl	display file with line numbers	nl file.txt
echo	print text or variables	echo "Hi"
printf	print formatted text	printf "Hi"
clear	clear the terminal screen	clear
history	Show command history	history
whoami	display current user name	whoami
who	Show logged-in users	who
id	display user and group IDs	id
hostname	Show system hostname	hostname

Here is a table of commands and their description:

command	Description	Syntax
date	Show current date	date
cal	Display calender	cal
uname	show operating system name	uname
file	identify file type	file file-name
Stat	display file information	Stat file_name
find	search for files	find file_name
Sort	Sort file contents	Sort file.txt
uniq	remove duplicate lines	uniq file.txt
wc	count lines, words and char	wc file.txt
diff	compare two files	diff file1 file2
cmp	compare files byte by byte	cmp file1 file2
split	split a file into parts	split file.txt
zip	compress file into zips	zip file.txt
unzip	extract zip files	unzip file.txt
kill	Kill process by name	kill PID
killall	Kill process by name	killall firefox
env	Show environment veriable	env
help	Show help for shell command	help cd
sudo	Run command as administrator	sudo command
expr	arithmatic operation	expr operand operator operand
grep	Search text in fils	grep "Hi" file.txt
df -h	show disk uses	df -h
df -sh	show folder size	df -sh folder
ps	show running process	ps
ping	Test network connection	ping google.com
ip addr	Show ip address	ip addr
bc	basic calculator operation	echo "Expression" | bc
reboot	Restart the system	reboot
shutdown	shut down the system	shut down now
poweroff	Turn off the system	poweroff
exit	exit terminal	exit

Conclusion:

This lab successfully introduced the basic commands of linux and their practical applications. The objectives of learning command line operations, file management, and system administration were achieved. The knowledge gained from this lab provides a strong foundation for advanced linux uses, shell scripting, software development and system administration.

References:

1. GNU Bash Manual
2. The linux command Line
3. Ubuntu Server Documentation